

Título do Trabalho

Matheus Borges

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1 Introduction

The continuous growth in global energy demand necessitates the optimization of hydrocarbon extraction from existing reservoirs. Following primary depletion, secondary recovery techniques, predominantly waterflooding, are implemented to maintain reservoir pressure and physically displace oil towards production wells [1]. This displacement fundamentally involves multiphase fluid flow within complex, heterogeneous porous media, where the dynamic interaction between immiscible phases governs the macroscopic recovery efficiency [2].

A critical operational limitation in these displacement processes arises when the displacing fluid (e.g., water) possesses a significantly lower dynamic viscosity than the displaced fluid (e.g., heavy crude oil). This adverse mobility ratio triggers a hydrodynamic instability at the moving fluid-fluid interface, recognized as the Saffman-Taylor instability, or viscous fingering [3]. Under such conditions, the planar interface becomes morphologically unstable, causing the less viscous injectant to channel through the oil phase in finger-like projections as illustrated in Figure 1.

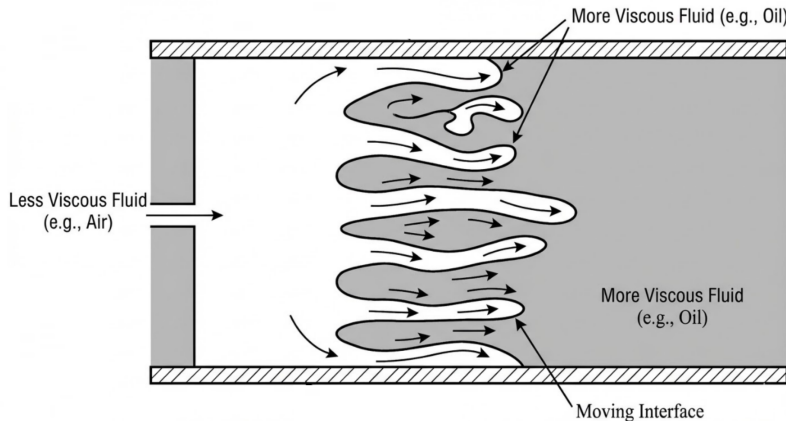


Figure 1: Fingers patterns example

In the context of Enhanced Oil Recovery (EOR), the Saffman-Taylor instability acts as a limiter to sweep efficiency. The propagation of viscous fingers leads to the premature breakthrough of the injected fluid at the production wells. Consequently, the water cut increases rapidly, leaving sub-

stantial volumes of bypassed, unrecoverable oil stranded within the reservoir matrix [4].

Various technological interventions have been developed to mitigate viscous fingering and improve the mobility ratio. Conventional chemical EOR strategies, such as polymer flooding, focus on increasing the apparent viscosity of the aqueous phase, thereby creating a more favorable mobility ratio and stabilizing the displacement front [5]. More recent advancements involve the injection of foams, surfactants, and nanoparticle-stabilized emulsions (smart fluids) designed to alter interfacial tension and modify local rheology [6]. However, these approaches often face limitations regarding thermal degradation, mechanical shear degradation in the near-wellbore region, complex reservoir retention, and significant operational costs [9, 10].

To address these limitations, this study investigates a novel active-control mechanism: the application of magnetic fluids (ferrofluids) as the displacing phase. By exposing the injected magnetic fluid to an external magnetic field, an artificial body force (the Kelvin force) is induced at the fluid interface. Theoretical models and preliminary experiments demonstrate that a properly aligned magnetic field can counteract the hydrodynamic destabilization forces, effectively suppressing the Saffman-Taylor fingering [7, 8].

Therefore, the primary objective of this thesis is the development and validation of a Computational Fluid Dynamics (CFD) simulation to model the multiphase displacement of oil by a magnetic fluid in a porous medium, supported by an analytical mathematical framework for theoretical validation. This computational framework will quantify the interfacial stabilization mechanisms and evaluate the sweep efficiency under varying magnetic field topologies and flow regimes.

2 Objectives

2.1 General Objective

The primary objective of this thesis is to evaluate the suppression of the Saffman-Taylor instability in porous media during Enhanced Oil Recovery (EOR) processes using magnetic fluids. This is achieved through the development of an integrated framework comprising a rigorous analytical mathematical formulation and a Computational Fluid Dynamics (CFD) simulation.

2.2 Specific Objectives

To accomplish this goals, the following specific objectives are established:

- **Analytical Formulation:** Formulate a Linear Stability Analysis (LSA) for the multiphase displacement involving a magnetic fluid under an external magnetic field to derive the theoretical stability criteria and the perturbation dispersion relations.
- **Computational Modeling:** Develop and verify a multiphase CFD model (e.g., Volume of Fluid or Phase Field) to simulate the nonlinear displacement of heavy oil by a magnetic fluid within a representative porous medium geometry.
- **Theoretical Validation:** Compare and validate the numerical predictions of the CFD model against the analytical thresholds established by the LSA, ensuring the physical accuracy of the computational framework prior to complex extrapolations.
- **Interfacial Dynamics Analysis:** Investigate the local fluid-fluid interaction, quantifying how varying magnetic field topologies physically alter the interfacial dynamics, either mitigating or potentially exacerbating finger growth.
- **Regime Mapping and EOR Evaluation:** Conduct a parametric study to map the flow regimes, comparing baseline conditions (conventional oil-water displacement) with magnetic fluid injection.

3 State of the Art

4 Theoretical Background

This section establishes the theoretical foundation necessary to comprehend, analytically formulate, and numerically simulate the active control of viscous fingering in porous media. It will start by examining fluid dynamics across different spatial scales, transitioning from the microscopic pore-level interactions to the macroscopic continuum descriptions governing flow in porous media. Subsequently, the physical principles of multiphase flow are addressed, with particular emphasis on the dynamic interactions between immiscible fluids, interfacial tension, and the resulting capillary phenomena. Building upon these interfacial concepts, the mathematical framework of Linear Stability Analysis (LSA) is introduced to rigorously describe the onset, growth, and governing parameters of the Saffman-Taylor instability.

Following the characterization of the instability, the physicochemical properties of magnetic fluids (ferrofluids) and the governing equations of magneto-hydrodynamics are presented. This provides the physical basis for how external magnetic fields exert stabilizing body forces (the Kelvin force) at the fluid-fluid interface. Finally, to bridge the gap between analytical stability criteria and complex, non-linear flow regimes, the theoretical basis of the Lattice Boltzmann Method (LBM) is detailed. Operating at a mesoscopic scale through the discrete Boltzmann equation, the LBM presents distinct advantages over traditional macroscopic Navier-Stokes solvers, particularly regarding the natural resolution of complex interfacial dynamics, phase separation, and the incorporation of external magnetic forces within the intricate geometries characteristic of porous matrixes.

4.1 Flow in Porous Media

4.1.1 Micro-Scale Porous Media: The Stokes Limit

At the microscopic scale, a porous medium is defined as a multiply connected domain $\Omega \subset \mathbb{R}^3$, partitioned into a fluid phase Ω_f and a stationary solid phase Ω_s . The interface $\Gamma = \partial\Omega_f \cap \partial\Omega_s$ represents the complex topological boundary of the pore walls [?].

In this domain, the flow physics is strictly governed by the ratio of inertial forces to viscous forces [?].

Dimensional Analysis and the Stokes Regime The characterization of the flow regime relies on the Reynolds number defined at the pore scale. Let ℓ be the characteristic length scale (e.g., hydraulic diameter of a pore throat) and U be the characteristic velocity scale within the pore.

$$Re_\ell = \frac{\rho U \ell}{\mu} \quad (1)$$

In micro-porous media (e.g., microfluidic networks, sedimentary rock matrices), ℓ is typically on the order of 10^{-6} to 10^{-4} m. Consequently, even for moderate pressure gradients, $Re_\ell \ll 1$.

Applying the non-dimensionalization variables $\mathbf{x}^* = \mathbf{x}/\ell$, $\mathbf{u}^* = \mathbf{u}/U$, and $p^* = p\ell/(\mu U)$, the steady-state Navier-Stokes momentum equation becomes:

$$Re_\ell(\mathbf{u}^* \cdot \nabla^*)\mathbf{u}^* = -\nabla^* p^* + \nabla^{*2}\mathbf{u}^* \quad (2)$$

As $Re_\ell \rightarrow 0$, the inertial term on the left-hand side vanishes. The governing physics reduces to a balance between pressure gradients and viscous diffusion.

Governing Micro-Scale Equations The fluid motion in Ω_f is described by the Stokes equations (creeping flow) [?]. The system is linear and instantaneous (time-independent):

- **Momentum Balance:**

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \implies \mu \nabla^2 \mathbf{u} - \nabla p = \mathbf{0} \quad (3)$$

- **Mass Conservation:**

$$\nabla \cdot \mathbf{u} = 0 \quad (4)$$

Here, $\boldsymbol{\sigma}$ is the Newtonian stress tensor:

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu\mathbf{D} \quad (5)$$

where $\mathbf{D} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the rate-of-strain tensor.

4.1.2 Meso-Scale Formulation: The Darcy-Brinkman Equation

The transition from the micro-scale Stokes regime to the meso-scale continuum is achieved by averaging the governing equations over a Representative Elementary Volume, V . Let V_f represent the fluid volume within V , and A_{fs} represent the interfacial area between the fluid and the solid particles.

We define the superficial average velocity, $\mathbf{v} \equiv \langle \mathbf{u} \rangle$, as:

$$\langle \mathbf{u} \rangle = \frac{1}{V} \int_{V_f} \mathbf{u} dV \quad (6)$$

Volume Averaging of the Stokes Momentum Equation Starting with the micro-scale stationary Stokes momentum equation:

$$-\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{b} = 0 \quad (7)$$

We apply the averaging operator $\langle \cdot \rangle$ to each term. The critical mathematical tool required is the *Spatial Averaging Theorem* (SAT), which allows us to commute the gradient and averaging operators [?]:

$$\langle \nabla \psi \rangle = \nabla \langle \psi \rangle + \frac{1}{V} \int_{A_{fs}} \psi \mathbf{n}_{fs} dA \quad (8)$$

where $\mathbf{n}_{fs}$ is the unit normal vector pointing from the fluid to the solid.

Derivation of Terms

1. **Pressure Gradient:** Applying the SAT to $-\nabla p$:

$$-\langle \nabla p \rangle = -\nabla \langle p \rangle^f - \frac{1}{V} \int_{A_{fs}} p \mathbf{n}_{fs} dA \quad (9)$$

Here, $\langle p \rangle^f$ is the intrinsic average pressure (pore pressure).

2. **Viscous Term (Brinkman & Drag):** Applying the SAT to $\mu \nabla^2 \mathbf{u}$ requires a second-order expansion. For high porosity media, we approximate [17]:

$$\mu \langle \nabla^2 \mathbf{u} \rangle \approx \underbrace{\mu_{eff} \nabla^2 \mathbf{v}}_{\text{Brinkman Term}} + \underbrace{\frac{1}{V} \int_{A_{fs}} \mu (\nabla \mathbf{u}) \cdot \mathbf{n}_{fs} dA}_{\text{Shear Drag}} \quad (10)$$

The first term represents the viscous diffusion of average momentum (important near boundaries), where μ_{eff} is the effective viscosity. The second term represents the viscous shear stress at the particle surfaces.

3. **Body Force:** Assuming $\rho\mathbf{b}$ is constant over the domain:

$$\langle\rho\mathbf{b}\rangle = \phi\rho\mathbf{b} \approx \rho\mathbf{b} \quad (\text{per unit fluid volume}) \quad (11)$$

The Meso-Scale Equation Combining these averaged terms, we group the surface integrals (pressure drag and viscous shear) into a single body force vector $\mathbf{f}$:

$$\mathbf{f} = \underbrace{-\frac{1}{V} \int_{A_{fs}} p \mathbf{n}_{fs} dA}_{\text{Form Drag}} + \underbrace{\frac{1}{V} \int_{A_{fs}} \mu (\nabla \mathbf{u}) \cdot \mathbf{n}_{fs} dA}_{\text{Viscous Drag}} \quad (12)$$

This term $\mathbf{f}$ represents the **average resistance force exerted by the fixed particles on the fluid**.

Substituting this back into the momentum balance yields the **Darcy-Brinkman equation**:

$$-\nabla\langle p\rangle^f + \mu_{eff}\nabla^2\mathbf{v} + \rho\mathbf{b} + \mathbf{f} = 0 \quad (13)$$

4.1.3 Macroscopic Limits: From Brinkman to Darcy

Having established the meso-scale momentum balance (Darcy-Brinkman), we can now formally construct Darcy's Law and determine the conditions under which each governing equation is valid. This process relies on the concept of **hydrodynamic screening**.

Construction of Darcy's Equation Starting from the generalized meso-scale equation derived previously (neglecting body forces $\mathbf{b}$ for brevity):

$$-\nabla p + \mu_{eff}\nabla^2\mathbf{v} - \frac{\mu}{K}\mathbf{v} = 0 \quad (14)$$

We perform a scaling analysis. Let L_c be the characteristic length scale of the macroscopic flow variation (e.g., the size of the domain or the scale of pressure gradients). The differential terms scale as:

$$\mu_{eff} \nabla^2 \mathbf{v} \sim \frac{\mu V}{L_c^2} \quad \text{and} \quad \frac{\mu}{K} \mathbf{v} \sim \frac{\mu V}{K} \quad (15)$$

The ratio of the viscous stress term (Brinkman) to the drag term (Darcy) is defined by the dimensionless **Darcy number** (Da):

$$\frac{\text{Viscous Stress}}{\text{Drag}} \sim \frac{\mu V / L_c^2}{\mu V / K} = \frac{K}{L_c^2} = Da \quad (16)$$

In the limit where the macroscopic length scale is very large compared to the pore structure ($L_c^2 \gg K$, or $Da \rightarrow 0$), the viscous stress term $\mu_{eff} \nabla^2 \mathbf{v}$ becomes negligible. The equation asymptotically reduces to the algebraic balance known as **Darcy's Law**:

$$-\nabla p - \frac{\mu}{K} \mathbf{v} = 0 \implies \mathbf{v} = -\frac{K}{\mu} \nabla p \quad (17)$$

4.1.4 Hydrodynamic Screening and Scale Selection

The transition between these regimes is governed by the **Brinkman screening length**, defined as $\lambda = \sqrt{K}$.

Physics of Screening: In a porous medium, the solid matrix acts as a momentum sink. Shear waves generated at a boundary (e.g., a wall with a no-slip condition) cannot propagate indefinitely into the bulk fluid; they are "screened" or damped by the Darcy drag within a distance of order $O(\sqrt{K})$.

This length scale allows us to select the appropriate governing equation based on the characteristic scale of the problem L_c relative to the screening length.

Regime Selection Guide

1. Micro-Scale Regime (Pore Level Physics)

Condition: The length scale L_c is comparable to the pore size ℓ_{pore} (or boundary layer δ inside a pore).

At this scale, the medium is not a continuum. We must solve the flow around individual solid obstacles. Drag is not a volume force but a boundary integration result.

$$L_c \sim \ell_{pore} \quad \rightarrow \quad -\nabla p + \mu \nabla^2 \mathbf{u} = 0 \quad (\text{Stokes Equation}) \quad (18)$$

2. Meso-Scale / Transition Regime (Boundary Layers)

Condition: The length scale L_c is comparable to the screening length $\sqrt{K}$.

This occurs near confining walls or interfaces between porous and pure fluid regions. The boundary layer thickness δ is of order $\sqrt{K}$. Here, both viscous diffusion (shear) and Darcy drag are equally important.

$$L_c \sim \sqrt{K} \quad \rightarrow \quad -\nabla p + \mu_{eff} \nabla^2 \mathbf{v} - \frac{\mu}{K} \mathbf{v} = 0 \quad (\text{Darcy-Brinkman}) \quad (19)$$

3. Macro-Scale Regime (Bulk Porous Flow)

Condition: The length scale L_c is much larger than the screening length.

The flow perceives the medium as a homogeneous resistance. Viscous shear is screened out, and the velocity is determined locally by the pressure gradient.

$$L_c \gg \sqrt{K} \quad \rightarrow \quad -\nabla p - \frac{\mu}{K} \mathbf{v} = 0 \quad (\text{Darcy Equation}) \quad (20)$$

4.1.5 Experimental Analogue: The Hele-Shaw Cell

To facilitate experimental validation and visualize the hydrodynamic instabilities discussed (e.g., Saffman-Taylor), we utilize the Hele-Shaw cell configuration. This setup serves as a rigorous two-dimensional analogue to Darcy flow in porous media [3].

Geometric Setup and Governing Equations Consider a fluid confined between two parallel plates separated by a small gap of thickness h , with lateral dimensions L and b such that $h \ll L, b$ [?]. The flow is driven by a pressure gradient ∇P in the (x, z) plane.

Under the conditions of low Reynolds number ($Re = \rho \bar{U} h / \mu \ll 1$) and high aspect ratio (Lubrication Approximation), the inertial terms in the Navier-Stokes equations are negligible [2]. The momentum balance reduces to a balance between the pressure gradient and the viscous shear across the gap thickness (y -direction):

$$\frac{\partial P}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2}, \quad \frac{\partial P}{\partial z} = \mu \frac{\partial^2 w}{\partial y^2} \quad (21)$$

where u and w are the velocity components in the x and z directions, respectively. Note that $\partial P / \partial y \approx 0$.

Velocity Profile and Averaging We apply no-slip boundary conditions at the walls ($y = 0$ and $y = h$):

$$u(0) = u(h) = 0 \quad (22)$$

Integrating the momentum equation twice with respect to y , we obtain the parabolic velocity profile characteristic of Poiseuille flow [?]:

$$\mathbf{u}(y) = \frac{1}{2\mu} \nabla P (y^2 - hy) \quad (23)$$

To establish the analogy with Darcy's Law, we calculate the gap-averaged velocity $\bar{\mathbf{u}}$ (or superficial velocity). This is obtained by integrating the profile over the gap height h :

$$\bar{\mathbf{u}} = \frac{1}{h} \int_0^h \mathbf{u}(y) dy = \frac{\nabla P}{2\mu h} \int_0^h (y^2 - hy) dy \quad (24)$$

Evaluating the definite integral $\int_0^h (y^2 - hy) dy = [\frac{y^3}{3} - \frac{hy^2}{2}]_0^h = -\frac{h^3}{6}$:

$$\bar{\mathbf{u}} = \frac{\nabla P}{2\mu h} \left(-\frac{h^3}{6} \right) = -\frac{h^2}{12\mu} \nabla P \quad (25)$$

Effective Permeability Definition Comparing this result with the standard Darcy's Law for a porous medium, $\bar{\mathbf{u}} = -\frac{K}{\mu} \nabla P$, we identify the effective permeability K of the Hele-Shaw cell [2]:

$$K_{HS} = \frac{h^2}{12} \quad (26)$$

This relationship demonstrates that the permeability in this experimental setup is strictly determined by the gap width h . By controlling h , we can precisely tune the "permeability" of the medium in the lab, enabling accurate scaling of instability phenomena such as viscous fingering.

4.2 Multiphase Flow in Porous Media

In the context of Enhanced Oil Recovery (EOR) and the specific problem of viscous fingering, the porous medium is rarely saturated by a single fluid. The simultaneous flow of two or more immiscible fluids (e.g., heavy oil and an aqueous magnetic fluid) requires an extension of the macroscopic equations derived for single-phase flow.

4.2.1 Phase Saturation and the Extended Darcy's Law

When multiple immiscible phases occupy the void space of a porous medium, the volume fraction of the pore space occupied by a specific phase α is defined as its saturation, S_α :

$$S_\alpha = \frac{V_\alpha}{V_p} \quad (27)$$

where V_α is the volume of phase α and V_p is the total pore volume. By definition, for a fully saturated medium containing n phases, $\sum_{i=1}^n S_i = 1$.

The presence of a second phase introduces capillary forces and reduces the available cross-sectional area for the flow of the first phase. Consequently, the absolute permeability K , which is strictly a geometric property of the solid matrix, is no longer sufficient to describe the flow conductance. We must introduce the concept of **effective permeability**, K_α , which is the permeability of the medium to phase α at a given saturation S_α .

The phenomenological extension of Darcy's Law for a multiphase system postulates that the superficial velocity of phase α , $\mathbf{v}_\alpha$, is driven by its own phase pressure gradient ∇p_α :

$$\mathbf{v}_\alpha = -\frac{K_\alpha}{\mu_\alpha}(\nabla p_\alpha - \rho_\alpha \mathbf{g}) \quad (28)$$

4.2.2 Relative Permeability

To normalize the flow equations, the effective permeability is expressed as the product of the absolute permeability K and a dimensionless scaling factor known as the **relative permeability**, $k_{r\alpha}$:

$$K_\alpha = K k_{r\alpha} \quad \text{where} \quad 0 \leq k_{r\alpha} \leq 1 \quad (29)$$

Unlike absolute permeability, $k_{r\alpha}$ is not solely a property of the rock. It is a highly non-linear, phenomenological function that depends primarily on the phase saturation ($k_{r\alpha} = f(S_\alpha)$), but also on the pore-size distribution, wettability of the solid matrix, and the flow history (hysteresis). The relative permeability accounts for the momentum dissipation at the fluid-fluid interfaces and the tortuous pathways forced upon one phase by the presence of the other.

Substituting into the extended Darcy's equation (neglecting gravity):

$$\mathbf{v}_\alpha = -\frac{K k_{r\alpha}}{\mu_\alpha} \nabla p_\alpha \quad (30)$$

4.2.3 Phase Mobility and the Mobility Ratio

The dynamic capacity of a specific phase to flow through the porous medium under a given pressure gradient is quantified by its **mobility**, λ_α . It couples the rock-fluid interaction parameter ($k_{r\alpha}$) with the intrinsic fluid property (dynamic viscosity μ_α):

$$\lambda_\alpha = \frac{K k_{r\alpha}}{\mu_\alpha} \implies \mathbf{v}_\alpha = -\lambda_\alpha \nabla p_\alpha \quad (31)$$

In a displacement process, such as the injection of a magnetic fluid (aqueous phase, subscript w) to displace heavy oil (subscript o), the hydrodynamic stability of the displacement front is governed by the **Mobility Ratio**, M . It is defined as the ratio of the mobility of the displacing fluid behind the front to the mobility of the displaced fluid ahead of the front:

$$M = \frac{\lambda_w}{\lambda_o} = \frac{k_{rw}/\mu_w}{k_{ro}/\mu_o} = \frac{k_{rw}\mu_o}{k_{ro}\mu_w} \quad (32)$$

The mobility ratio is the critical dimensionless parameter determining macroscopic sweep efficiency:

- $M \leq 1$ (**Favorable Displacement**): The displaced fluid is more mobile than the injected fluid. The pressure field is stable, and the displacement front tends to remain planar and uniform.
- $M > 1$ (**Adverse Displacement**): The injected fluid has higher mobility (e.g., injecting low-viscosity water to displace high-viscosity oil). The displacement front becomes inherently unstable to infinitesimally small perturbations.

This adverse mobility condition ($M > 1$) is the fundamental physical trigger for the hydrodynamic instability known as viscous fingering, which necessitates the introduction of active control mechanisms such as magnetic fluids.

4.3 Phase-Field Modeling: The Cahn-Hilliard Equation

The mobility ratio criterion established above identifies when an immiscible displacement becomes hydrodynamically unstable, but it does not by itself

prescribe how the fluid-fluid interface is to be represented and tracked through space and time. In a classical sharp-interface formulation, the interface $\Gamma(t)$ is treated as a co-dimension-one manifold whose evolution is governed by free-boundary kinematic and dynamic conditions (continuity of normal velocity and a Laplace-type pressure jump set by the local curvature and the surface tension). Solving such a formulation in geometries as topologically complex as a porous matrix, where the interface routinely undergoes pinch-off, coalescence, and fingering events, is analytically intractable and numerically requires explicit reconstruction of $\Gamma(t)$ at every time step.

To circumvent these difficulties we adopt a diffuse-interface (phase-field) description, in which the interface is regularised over a finite thickness W by introducing a conserved scalar order parameter $\phi(\mathbf{x}, t)$ that varies smoothly between two bulk equilibrium values:

$$\phi(\mathbf{x}, t) = \begin{cases} +1 & \text{invading (magnetic) phase,} \\ -1 & \text{resident (oleic) phase,} \\ \in (-1, +1) & \text{interfacial region.} \end{cases} \quad (33)$$

The geometric interface is then implicitly defined as the level set $\{\mathbf{x} : \phi(\mathbf{x}, t) = 0\}$, and topological events are handled automatically without remeshing or geometric reconstruction.

4.3.1 Ginzburg-Landau Free-Energy Functional

The thermodynamics of the two-phase system is encoded by the Ginzburg-Landau free-energy functional [15]:

$$\mathcal{F}[\phi] = \int_{\Omega} \left[\underbrace{\beta(\phi^2 - 1)^2}_{\text{double-well (bulk)}} + \underbrace{\frac{\kappa}{2} |\nabla \phi|^2}_{\text{gradient penalty}} \right] d\Omega. \quad (34)$$

The bulk term $\beta(\phi^2 - 1)^2$ is a symmetric double-well potential with degenerate minima at $\phi = \pm 1$, expressing the thermodynamic preference of the system for pure phases over mixed states. The gradient term $(\kappa/2)|\nabla \phi|^2$ penalises sharp variations of ϕ and is responsible for the finite interface thickness; without it, minimisation of $\mathcal{F}$ would drive the interface towards a singular discontinuity. The two parameters β and κ are independent, with positive units, and jointly control the macroscopic interfacial properties of the diffuse model.

4.3.2 Chemical Potential and the Cahn-Hilliard Equation

The chemical potential associated with the order parameter is defined as the functional derivative of $\mathcal{F}$:

$$\mu_c = \frac{\delta \mathcal{F}}{\delta \phi} = 4\beta \phi(\phi^2 - 1) - \kappa \nabla^2 \phi. \quad (35)$$

For an incompressible two-phase flow with velocity field $\mathbf{u}$, the conservative evolution of the order parameter is governed by the advective Cahn-Hilliard equation [15]:

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = M \nabla^2 \mu_c, \quad (36)$$

where M is the mobility coefficient (with units of $L^2 T^{-1}$ per unit chemical potential). The advective term on the left transports the phase distribution by the hydrodynamic field, while the right-hand side drives the diffusive relaxation of ϕ towards the local minimisers of $\mathcal{F}$, counteracting the numerical thickening of the interface caused by advection. Because the right-hand side can be written as the divergence of a flux,

$$M \nabla^2 \mu_c = \nabla \cdot (M \nabla \mu_c) \implies \mathbf{J} = \phi \mathbf{u} - M \nabla \mu_c, \quad \frac{\partial \phi}{\partial t} = -\nabla \cdot \mathbf{J}, \quad (37)$$

the Cahn-Hilliard equation conserves the total amount of each phase exactly in closed domains:

$$\frac{d}{dt} \int_{\Omega} \phi d\Omega = - \oint_{\partial\Omega} \mathbf{J} \cdot \hat{\mathbf{n}} dS. \quad (38)$$

This conservation property distinguishes the Cahn-Hilliard model (fourth-order in ϕ , conservative) from the Allen-Cahn model (second-order, non-conservative) and is essential for long-time simulations of immiscible displacement, where loss of mass would bias the measured fingering growth rates and saturation profiles.

4.3.3 Equilibrium Profile and Interfacial Tension

For a stationary planar interface aligned with the $z = 0$ plane (no advection, $\partial_t \phi = 0$), the Cahn-Hilliard equation reduces to $\nabla^2 \mu_c = 0$ subject to $\phi(\pm\infty) = \pm 1$ and $\mu_c(\pm\infty) = 0$, yielding $\mu_c \equiv 0$ throughout the domain. The chemical-potential definition then becomes a nonlinear ordinary differential equation,

$$4\beta \phi(\phi^2 - 1) - \kappa \frac{d^2 \phi}{dz^2} = 0, \quad (39)$$

whose physical solution is the hyperbolic-tangent profile

$$\phi_{eq}(z) = \tanh\left(\frac{z\sqrt{2}}{W}\right), \quad W = \sqrt{\frac{\kappa}{2\beta}}, \quad (40)$$

where W is the characteristic interface thickness. Integrating the gradient-energy density across the interface yields the effective interfacial tension associated with the diffuse-interface model:

$$\sigma = \int_{-\infty}^{+\infty} \kappa \left(\frac{d\phi_{eq}}{dz}\right)^2 dz = \frac{2\sqrt{2}}{3} \sqrt{\kappa\beta}. \quad (41)$$

Inverting these two algebraic identities permits independent calibration: given target physical values of σ and W , the model parameters are recovered as $\beta = 3\sqrt{2}\sigma/(4W)$ and $\kappa = 3\sqrt{2}\sigma W/4$. This separation of the two macroscopic interface properties into two independent model parameters is one of the key reasons for choosing the Cahn-Hilliard formulation as the multiphase backbone in this work.

4.3.4 Coupling to the Momentum Equation: Korteweg Stress

To close the multiphase Navier-Stokes/Darcy-Brinkman momentum balance, the diffuse-interface model must supply the capillary body force. By Noether's theorem applied to translations of the free energy $\mathcal{F}$, the surface-tension contribution can be written in compact form as the Korteweg stress, whose body-force representation is

$$\mathbf{F}_s = \mu_c \nabla \phi. \quad (42)$$

Outside the diffuse interfacial layer $\nabla \phi \approx 0$ and $\mathbf{F}_s$ vanishes identically; within the layer the force is non-zero and acts along the local interface normal, with a magnitude that reproduces, in the sharp-interface limit ($W \rightarrow 0$ at fixed σ), the standard capillary forcing $\sigma \kappa_c \hat{\mathbf{n}} \delta_\Gamma$, where κ_c is the local mean curvature. In this manner the Cahn-Hilliard equation supplies, in a single scalar field ϕ , three quantities required by the full multiphase formulation: the implicit geometric tracking of the interface (level set $\phi = 0$), the conservation of phase volumes (divergence form of the flux $\mathbf{J}$), and the capillary forcing $\mathbf{F}_s$ to be inserted into the momentum equation. The numerical implementation of this coupling within the Lattice Boltzmann framework is detailed in the next chapter.

4.4 Magnetic Fluids: Fundamentals and Properties

To effectively control hydrodynamic instabilities such as viscous fingering, this study employs magnetic fluids as the active displacing phase. Magnetic fluids, commonly referred to as ferrofluids, are a unique class of smart fluids that exhibit magnetic properties. As described by Rosensweig [?], these fluids are not true homogenous solutions, but rather highly stable colloidal suspensions.

A typical ferrofluid consists of three primary components:

1. **Magnetic Nanoparticles:** Single-domain particles, typically composed of magnetite (Fe_3O_4) or maghemite ($\gamma\text{-Fe}_2\text{O}_3$), with diameters on the order of 10 nanometers.
2. **Carrier Liquid:** The continuous non-magnetic fluid phase in which the particles are dispersed, which can be aqueous or organic (e.g., water, kerosene, or synthetic oils).
3. **Surfactant Coating:** To prevent the agglomeration of the nanoparticles due to strong short-range van der Waals attractive forces and magnetic dipole-dipole interactions, the particles are coated with a layer of long-chain surfactant molecules (steric repulsion) or stabilized via electrostatic charge.

A defining characteristic of these stable suspensions is their superparamagnetic behavior. In the absence of an external magnetic field, the magnetic moments of the individual nanoparticles are randomly oriented due to thermal agitation, resulting in a net macroscopic magnetization of zero. Upon the application of an external magnetic field, the particle moments rapidly align with the field lines.

The mathematical validity of treating this biphasic suspension as a homogeneous, single-phase continuum in macroscopic flow equations relies on two critical scale separations: spatial and temporal [?].

Spatially, let d_p be the average diameter of the coated magnetic nanoparticles and D_p be the characteristic length scale of the porous matrix (e.g., the pore throat diameter). As established by Rosensweig [?], typical ferrofluid nanoparticles possess $d_p \sim \mathcal{O}(10^{-8} \text{ m})$. Conversely, extensive petrophysical characterizations by Dullien [?] indicate that standard reservoir rock pore throats are on the order of $D_p \sim \mathcal{O}(10^{-5} \text{ to } 10^{-4} \text{ m})$. The spatial scale ratio

is therefore:

$$\epsilon = \frac{d_p}{D_p} \sim 10^{-4} \ll 1 \quad (43)$$

Because $\epsilon \ll 1$, the discrete nature of the nanoparticles is negligible, and the fluid perfectly conforms to the continuum hypothesis required for the Navier-Stokes and Darcy formulations.

Temporally, the fluid's response to dynamic flow conditions and changing magnetic fields is governed by its magnetic relaxation time, τ . In ferrofluids, relaxation occurs via two parallel mechanisms formulated by Rosensweig [?]: the physical rotation of the particle within the fluid (Brownian relaxation, τ_B) and the internal rotation of the magnetic moment within the particle's crystalline lattice (Néel relaxation, τ_N). The effective relaxation time is dominated by the faster of the two processes:

$$\tau = \left(\frac{1}{\tau_B} + \frac{1}{\tau_N} \right)^{-1} \quad (44)$$

For 10 nm magnetite particles, τ is extremely short, typically $\mathcal{O}(10^{-6} \text{ s})$. Conversely, the characteristic hydrodynamic time scale of the flow in the porous medium is $t_f = D_p/U$, which is generally $\mathcal{O}(10^{-2} \text{ to } 1 \text{ s})$.

This temporal separation is rigorously quantified by the dimensionless Deborah number (De), a standard rheological metric defined as the ratio of the fluid's relaxation time to the characteristic time of the flow:

$$De = \frac{\tau}{t_f} \ll 1 \quad (45)$$

Because De is asymptotically close to zero, the magnetic moments adjust to local field variations and flow vorticity instantaneously relative to the fluid motion. This mathematical condition ensures that the ferrofluid lacks visco-elastic memory, behaving as a purely viscous Newtonian fluid, and that its macroscopic magnetization $\mathbf{M}$ is always in collinear thermodynamic equilibrium with the applied field $\mathbf{H}$ (i.e., $\mathbf{M} = \chi\mathbf{H}$). Consequently, these spatial and temporal conditions provide the exact physical foundation to dynamically control the fluid interface in porous media via macroscopic external fields.

4.5 The Kelvin Force and Effective Magnetic Pressure

The physical coupling between the external magnetic field and the fluid flow within the porous medium occurs through a volumetric body force known

as the Kelvin force. Because the ferrofluid is electrically non-conductive, the standard Lorentz force (associated with moving free charges) is completely absent. The macroscopic magnetic force density $\mathbf{f}_m$ acting on a magnetizable fluid is defined as:

$$\mathbf{f}_m = \mu_0(\mathbf{M} \cdot \nabla)\mathbf{H} \quad (46)$$

where $\mathbf{H}$ is the macroscopic magnetic field intensity, $\mathbf{M}$ is the fluid's magnetization, and μ_0 is the vacuum magnetic permeability.

Under the condition of a superparamagnetic fluid in thermodynamic equilibrium, the magnetization is collinear and proportional to the magnetic field, such that $\mathbf{M} = \chi\mathbf{H}$, where χ is the magnetic susceptibility. Applying vector calculus identities, and given that the magnetic field is irrotational in the absence of free currents ($\nabla \times \mathbf{H} = \mathbf{0}$), this force can be analytically rewritten as the gradient of the squared magnetic field intensity:

$$\mathbf{f}_m = \frac{1}{2}\mu_0\chi\nabla H^2 \quad (47)$$

As extensively detailed in the comprehensive review of ferrohydrodynamics by Gontijo and Cunha [?], because this Kelvin force is strictly conservative ($\nabla \times \mathbf{f}_m = \mathbf{0}$), it acts mathematically as an artificial pressure gradient. Incorporating this volumetric body force into the macroscopic Darcy's law yields the coupled momentum equation for the magnetic phase:

$$\frac{\eta}{K}\mathbf{v} = -\nabla p + \nabla \left(\frac{1}{2}\mu_0\chi H^2 \right) \quad (48)$$

Because both the thermodynamic pressure gradient and the Kelvin force are gradients of scalar fields, the magnetic force can be analytically absorbed into a modified, or effective, pressure p^* :

$$p^* = p - \frac{1}{2}\mu_0\chi H^2 \implies \frac{\eta}{K}\mathbf{v} = -\nabla p^* \quad (49)$$

This fundamental mathematical arrangement demonstrates that the external magnetic field locally modulates the pressure field of the fluid. Furthermore, since the velocity vector $\mathbf{v}$ is strictly proportional to the gradient of this effective pressure (p^*), the velocity field remains irrotational. This property permits the mathematical definition of a velocity potential ϕ such that $\mathbf{v} = \nabla\phi$, fulfilling Laplace's equation in incompressible domains. The structural equivalence between the Kelvin force and a pressure drop establishes the

theoretical mechanism by which the external magnetic field can be precisely manipulated to alter local velocity gradients at the interface, counteracting viscous fingering.

5 Linear Stability Analysis of Interfaces in Porous Media

This chapter develops the asymptotic analysis and linear stability modeling for the interface between two immiscible fluids in a porous medium (or equivalently, in a Hele-Shaw cell). The formulation is structured in two parts. The first encompasses classical hydrodynamic instabilities driven by viscous gradients and capillary forces (Saffman-Taylor instability), and the second addresses the instability interaction when an external magnetic field is applied to a ferrofluid.

1. Physical Description of the System and Nomenclature

The macroscopic system consists of two viscous and incompressible fluids confined in a homogeneous and isotropic porous medium. A Cartesian coordinate system is defined wherein the y -direction is orthogonal to the planar wave interface.

Fluid 1 (the invading fluid) is located in the lower region ($y < 0$) and is injected with a uniform filtration velocity $\frac{V}{\epsilon}$ in the positive vertical direction ($\hat{j}$). In the first case, it behaves as a standard Newtonian fluid, whereas in the second case, it is a ferrofluid; in both scenarios, it displaces **Fluid 2** (the defending fluid), which is Newtonian and situated in the upper region ($y > 0$). The system is subject to gravitational acceleration acting in the $-y$ direction. This physical configuration is depicted in the figure below.

The interface position, in both cases, is defined by Equation 50:

$$y(x, t) = \frac{V}{\epsilon}t + \lambda(x, t) \quad (50)$$

Here, y denotes the vertical position, $\frac{V}{\epsilon}t$ is the mean position of the interface moving with constant velocity, and $\lambda(x, t)$ represents the perturbation (distortion) at the interface.

The formulation relies on the macroscopic sharp-interface approximation. This theoretical idealization is valid under the assumption of high Péclet numbers ($Pe \gg 1$), where convective transport dominates molecular diffusion, thereby suppressing dynamic mixing between the immiscible phases and confining interfacial gradients to a negligibly thin region.

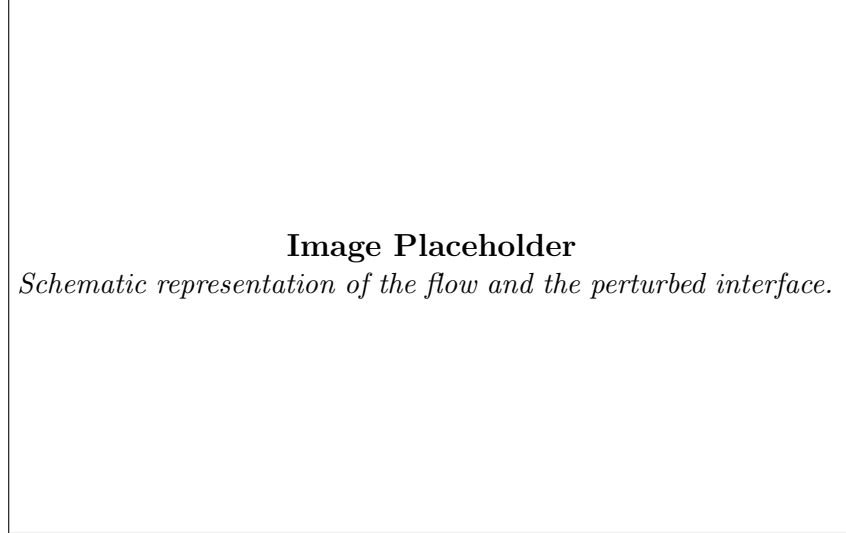


Figura 2: Schematic representation of the flow and the perturbed interface.

5.1 Development of the Kinematic Equation

The temporal evolution of the boundary between the fluids is governed by the kinematic boundary condition. Physically, this formulation establishes that the interface acts as a continuous material surface; that is, there is no mass flux (or fluid crossover) through it. This ensures that any fluid particle originally located at the interface remains constrained to this geometric boundary throughout the flow. To mathematically model this restriction, the position of the moving boundary is implicitly described by a scalar function $F(x, y, t) = 0$. Considering the system in a stationary reference frame, where the base state of the (planar) interface is displaced with a constant mean velocity $\frac{V}{\epsilon}$ in the vertical y -direction, and upon which a perturbation denoted by $\lambda(x, t)$ is superimposed, the surface topology is rigorously defined by Equation 51.

$$F(x, y, t) = y - \left(\frac{Vt}{\epsilon} + \lambda(x, t) \right) = 0 \quad (51)$$

The strict requirement that a fluid element does not leave the interface implies, under the Lagrangian formulation, that the rate of change of F following the fluid particle's motion must be identically zero. Consequently, the material (or convective) derivative of the surface function vanishes at the

boundary, ensuring that the geometric propagation velocity of the interface is perfectly coupled with the local velocity of the flow field $\mathbf{v}$. This fundamental condition is expressed by Equation 52.

$$\frac{DF}{Dt} = \frac{\partial F}{\partial t} + \mathbf{v} \cdot \nabla F = 0 \quad (52)$$

Restricting the formulation to a two-dimensional flow with velocity field $\mathbf{v} = (v_x, v_y)$, the expansion of the material derivative operator requires the explicit evaluation of the temporal and spatial partial derivatives of $F(x, y, t)$. Following these procedures, the vertical component of the fluid velocity (v_y) at the boundary can be evaluated by Equation 53.

$$v_y = \frac{\partial \lambda}{\partial t} + v_x \frac{\partial \lambda}{\partial x} + \frac{V}{\epsilon} \quad (53)$$

According to Darcy's Law, the velocity vector is directly proportional to the pressure gradient ($\mathbf{v} \propto \nabla p$). As the curl operator applied to any gradient field is identically zero ($\nabla \times \nabla p = \mathbf{0}$), irrotationality is intrinsic to the system, ensuring the existence of a scalar potential ϕ that describes the velocity field ($\mathbf{v} = \nabla \phi$). Applying this to Equation 53 consolidates a kinematic boundary partial differential equation, valid for the interface between the fluids, as described by Equation 54.

$$\frac{\partial \lambda}{\partial t} + \frac{\partial \phi}{\partial x} \frac{\partial \lambda}{\partial x} + \frac{V}{\epsilon} = \frac{\partial \phi}{\partial y} \quad (54)$$

5.2 Equilibrium Solution

5.3 Linearization of the Kinematic Equation

5.3.1 Velocity Potential Perturbation

To investigate the evolution of instabilities at the boundary, a linear perturbation analysis is applied. The total velocity potential field is decomposed into the superposition of a base state ϕ^0 , which represents the exact macroscopic solution, and an infinitesimal spatiotemporal fluctuation ϕ' , as shown in Equation 55:

$$\phi(x, y, t) = \phi^0(y, t) + \phi'(x, y, t) \quad (55)$$

The base state corresponds to the geometric condition of a planar interface undergoing uniform translation, characterized by a constant one-dimensional flow with velocity $\frac{V}{\epsilon}$ in the y -direction. Consequently, the vertical component of the base potential gradient is given by Equation 56:

$$\frac{\partial \phi^0}{\partial y} = \frac{V}{\epsilon} \implies \phi^0 = \frac{V}{\epsilon}y + f(t) \quad (56)$$

The incompressibility of the fluids ($\nabla \cdot \mathbf{v} = 0$) requires the perturbation potential to satisfy Laplace's equation in both domains. Physically, since the hydrodynamic fluctuations are induced strictly by the local instability of the interface, the mathematical condition of asymptotic decay at infinity is imposed. Assuming Fluid 1 in the lower domain and Fluid 2 in the upper domain, it is formulated as Equations 57 and 58:

$$\nabla^2 \phi'_1 = 0 \quad \text{and} \quad \nabla^2 \phi'_2 = 0 \quad (57)$$

$$\phi'_1 \rightarrow 0 \text{ as } y \rightarrow -\infty, \quad \phi'_2 \rightarrow 0 \text{ as } y \rightarrow \infty \quad (58)$$

5.3.2 Linearization of the Kinematic Condition

The linearization proceeds by inserting the decomposed potential into the general kinematic equation. Under the assumption of infinitesimal perturbations, products between fluctuating terms (e.g., $\frac{\partial \phi'}{\partial x} \frac{\partial \lambda}{\partial x}$) constitute higher-order quantities and are thus neglected. Evaluating this expression at the base state interface position ($y \approx Vt/\epsilon$), the terms governing the constant translation of the reference frame cancel out. Consequently, the resulting linearized kinematic condition directly correlates the temporal growth rate of the perturbation with the vertical component of the perturbed velocity field, presented in Equation 59:

$$\frac{\partial \lambda}{\partial t} = \frac{\partial \phi'}{\partial y} \quad (59)$$

The linear stability analysis of the interface proceeds through the decomposition of perturbations into normal modes. The wavenumber α , which characterizes the spatial periodicity, and the complex growth rate s are introduced in Equation 60:

$$\alpha = \frac{2\pi}{\lambda}, \quad s = \zeta - i\omega \quad (60)$$

where ζ quantifies the exponential amplification factor (stability criterion) and ω represents the temporal oscillatory component. An Ansatz is postulated for the fluctuations in the form of two-dimensional harmonic waves, formulated in Equation 61:

$$\begin{aligned}\lambda(x, t) &= \lambda_0 e^{i\alpha x + st} \\ \phi'_1(x, y, t) &= A_1(y) e^{i\alpha x + st} \quad (y < y_0) \\ \phi'_2(x, y, t) &= A_2(y) e^{i\alpha x + st} \quad (y > y_0)\end{aligned}\tag{61}$$

Inserting these modal solutions into Laplace's equation ($\nabla^2 \phi' = 0$) converts the Partial Differential Equation into a second-order Ordinary Differential Equation (ODE) for the vertical amplitude, given by Equation 62:

$$\frac{d^2 A_j}{dy^2} - \alpha^2 A_j = 0\tag{62}$$

By imposing the physical requirement of asymptotic decay of the perturbations at the far-field limit from the boundary ($\phi' \rightarrow 0$ as $y \rightarrow \pm\infty$), only the bounded solutions are selected in Equation 63:

$$A_1(y) = C_1 e^{+\alpha(y-y_0)}, \quad A_2(y) = C_2 e^{-\alpha(y-y_0)}\tag{63}$$

5.3.3 Application of the Kinematic Condition

Closure of the system requires determining the integration constants C_1 and C_2 via the linearized kinematic condition ($\frac{\partial \lambda}{\partial t} = \frac{\partial \phi'}{\partial y}$) evaluated at the unperturbed interface $y = y_0$. Under the normal mode representation, differential operations translate into the algebraic mapping $\partial_t \rightarrow s$ and $\partial_y \rightarrow \pm\alpha$. Applying these relations to the boundary condition yields the integration constants as functions of the growth rate and wavenumber. Consequently, the perturbed potential fields become uniquely coupled to the amplitude of the geometric deformation λ_0 , yielding Equation 64:

$$\begin{aligned}\phi'_1 &= \frac{s}{\alpha} \lambda_0 e^{+\alpha(y-y_0)} e^{i\alpha x + st} \\ \phi'_2 &= -\frac{s}{\alpha} \lambda_0 e^{-\alpha(y-y_0)} e^{i\alpha x + st}\end{aligned}\tag{64}$$

5.4 Dynamic Boundary Condition for the Non-Magnetic Case

The pressure in each fluid is governed by the extended Darcy's Law (accounting for gravitational potential). The relationship between pressure and potential for fluids 1 and 2 is defined in Equations 65 and 66:

$$p_1 = -\frac{\mu_1}{k_1}\phi_1 + \rho_1 \mathbf{g} \cdot \mathbf{x} \quad (65)$$

$$p_2 = -\frac{\mu_2}{k_2}\phi_2 + \rho_2 \mathbf{g} \cdot \mathbf{x} \quad (66)$$

The dynamic boundary condition requires the pressure jump across the interface to be balanced by surface tension, as stated in Equation 67:

$$p_2 - p_1 = \sigma c \quad (67)$$

where the linearized curvature c is given by $c \approx \frac{\partial^2 \lambda}{\partial x^2}$. Substituting the pressure expressions evaluated at the perturbed interface $y = Vt/\epsilon + \lambda$ into the boundary condition yields Equation 68:

$$\frac{\mu_1}{k_1}\phi_1 - \frac{\mu_2}{k_2}\phi_2 - (\rho_2 - \rho_1)g\lambda - (\rho_2 - \rho_1)g\frac{Vt}{\epsilon} = \sigma \frac{\partial^2 \lambda}{\partial x^2} \quad (68)$$

Linearization of the Dynamic Condition

In Eq. 68, the zeroth-order terms (hydrostatic equilibrium and base flow) cancel out, yielding the perturbation equation expressed in Equation 69:

$$\frac{\mu_1}{k_1}\phi'_1 - \frac{\mu_2}{k_2}\phi'_2 - (\rho_2 - \rho_1)g\lambda = \sigma \frac{\partial^2 \lambda}{\partial x^2} \quad (69)$$

Application of the Dynamic Condition and Dispersion Relation

The linearized dynamic condition is given by Equation 70:

$$\frac{\mu_1}{k_1}\phi'_1 - \frac{\mu_2}{k_2}\phi'_2 + \left[-g(\rho_2 - \rho_1) + \frac{V}{\epsilon} \left(\frac{\mu_1}{k_1} - \frac{\mu_2}{k_2} \right) \right] \lambda = \sigma \frac{\partial^2 \lambda}{\partial x^2} \quad (70)$$

Inserting the modal solutions for ϕ'_1 , ϕ'_2 , and λ into this condition, and factoring out the common harmonic dependence, reduces the differential equation to an algebraic dispersion relation. Solving for the complex growth rate $s = \zeta - i\omega$ yields Equation 71:

$$\zeta - i\omega = \frac{\alpha}{\frac{\mu_1}{k_1} + \frac{\mu_2}{k_2}} \left[\frac{V}{\epsilon} \left(\frac{\mu_2}{k_2} - \frac{\mu_1}{k_1} \right) + g(\rho_2 - \rho_1) - \sigma\alpha^2 \right] \quad (71)$$

5.4.1 Analysis of the Real and Imaginary Parts

Since all terms on the right-hand side of Eq. 71 are purely real (viscosity, permeability, density, and surface tension), it follows:

- **Imaginary Part (ω):** As shown in Equation 72:

$$\omega = 0 \quad (72)$$

This indicates the principle of exchange of stabilities holds. The perturbation is non-oscillatory; it grows or decays monotonically. This strictly monotonic evolution reflects the highly dissipative nature of flow in porous media. In the Darcy regime (creeping flow, $Re \ll 1$), inertial forces are entirely negligible compared to viscous drag from the porous matrix. Consequently, the system lacks the kinetic energy storage mechanism required to sustain inertial oscillations or traveling waves, behaving analogously to a strongly overdamped oscillator.

- **Real Part (ζ):** Expressed in Equation 73:

$$\zeta = \frac{\alpha}{\frac{\mu_1}{k_1} + \frac{\mu_2}{k_2}} \left[\underbrace{\frac{V}{\epsilon} \left(\frac{\mu_2}{k_2} - \frac{\mu_1}{k_1} \right)}_{\text{Viscous Forcing}} + \underbrace{g(\rho_2 - \rho_1)}_{\text{Gravity}} - \underbrace{\sigma\alpha^2}_{\text{Capillarity}} \right] \quad (73)$$

5.4.2 Nondimensionalization

To construct a generalized parametric space, the system is nondimensionalized. The characteristic velocity scale is defined by the uniform filtration velocity $U = V/\epsilon$. The characteristic length scale b is selected based on the specific geometry, representing either the macroscopic characteristic length of the porous domain or the gap thickness in the Hele-Shaw cell analog. This physical selection establishes the convective timescale $T = b/U$.

Defining the characteristic length scale b , velocity U , and time $T = b/U$, the dimensionless variables are defined in Equation 74:

$$\zeta^* = \zeta \frac{b}{U}, \quad \alpha^* = \alpha b \quad (74)$$

Assuming a homogeneous medium ($k_1 = k_2 = k$) and defining the viscosity contrast $A = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}$ (viscous Atwood number) and $M = \mu_2/\mu_1$, the equation can be rewritten using the following dimensionless numbers:

- $Ca = \frac{\mu U}{\sigma}$ (Capillary Number)
- $Bo = \frac{\Delta \rho g b^2}{\sigma}$ (Bond Number)
- Da (Darcy Number, implicit in the mobility terms)

The final dimensionless dispersion relation becomes Equation 75:

$$\zeta^*(\alpha^*) = \alpha^* \frac{M-1}{M+1} + \frac{Da}{Ca(1+M)} \alpha^* (Bo - \alpha^{*2}) \quad (75)$$

5.4.3 Graphical analysis of the equation

The figure 3 shows the dispersion relation plot obtained from the linear stability analysis governing the system described in equation 75. The dynamic stability of the interface is determined by the sign of the dimensionless growth rate, ζ^* . Configurations with $\zeta^* > 0$ indicate amplification of interfacial perturbations in time, which characterizes the unstable regime. In contrast, the semi plane $\zeta^* < 0$ defines the asymptotically stable region, where fluctuations are strictly damped. It is also observed that, for large dimensionless wavenumbers ($\alpha^* \gg 1$), the curve shows a sharp parabolic decay toward negative infinity ($\zeta^* \rightarrow -\infty$). Analytically, this behavior is governed by the dominance of the negative third-order polynomial term ($-\alpha^{*3}$) in the dispersion relation (Equation 75). Physically, this decay reflects the restraining effect of surface tension: at microscopic length scales (high α^*), the increase in interfacial curvature generates extreme capillary pressure gradients that act as a dominant restoring force, rapidly suppressing the growth of short waves and ensuring the stabilization of the system in this limit.

The figure 4 illustrates the individual contributions of the physical mechanisms governing the interfacial stability. The total dimensionless growth rate (ζ^*) is the linear superposition of viscous, gravitational, and capillary effects.

As shown in the plot, the viscous forcing and gravity terms exhibit a linear dependence on the dimensionless wavenumber (α^*). Under the evaluated unstable conditions ($M > 1$ and $Bo > 0$), both act as purely destabilizing forces. If acting alone, they would promote continuous and unbounded growth of perturbations across all spatial scales.

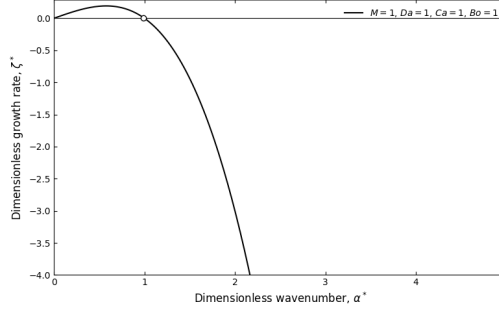


Figura 3: Enter Caption

Conversely, the capillarity term provides the essential stabilizing mechanism. It scales with the negative cube of the wavenumber ($-\alpha^{*3}$). The graph clearly demonstrates the physical interplay between these forces across the spectrum:

- Long-wave regime (low α^*): The linear destabilizing forces (viscosity and gravity) dominate, causing the total growth rate to increase and driving the macroscopic instability of the interface.
- Short-wave regime (high α^*): As the wavenumber increases, the wavelength decreases, resulting in a highly curved interface. In this region, the cubic capillary term rapidly outpaces the linear components.

This dynamic explains the characteristic parabolic shape of the total growth rate curve. The system reaches a peak of maximum instability before undergoing a sharp decline. Exactly where the capillary damping equals the combined viscous and gravitational forces, the total curve crosses the horizontal axis ($\zeta^* = 0$). Beyond this cut-off wavenumber, the surface tension acts as a strict low-pass filter, completely suppressing microscopic perturbations.

5.5 Dynamic Boundary Condition for the Magnetic Case

The introduction of an external magnetic field modifies the stress balance at the interface due to the polarization of the ferrofluid (Fluid 1). To prevent notational ambiguity with the dynamic viscosity (μ), the absolute magnetic permeability of the fluids is defined herein as μ_{m1} and μ_{m2} . Since Fluid 2 is

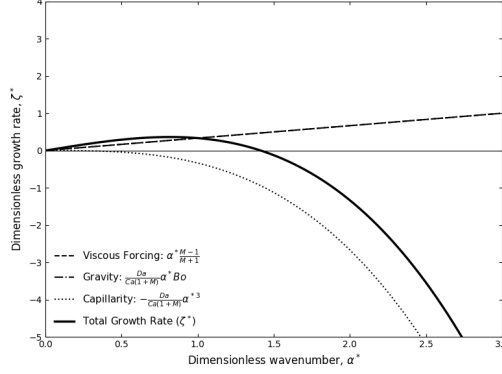


Figure 4: Enter Caption

an ordinary, non-magnetic Newtonian fluid, its permeability equals that of the vacuum ($\mu_{m2} = \mu_0$).

5.5.1 Extended Darcy's Law and Pressure Jump

The momentum equation for the ferrofluid is governed by the extended Darcy's law, which incorporates both the gravitational potential and the Kelvin body force arising from the gradient of the magnetic energy:

$$\frac{\mu_1}{k} \mathbf{v}_1 = -\nabla p_1 + \rho_1 \mathbf{g} + \mu_0 \chi_1 \nabla \left(\frac{1}{2} H^2 \right) \quad (76)$$

where χ_1 is the magnetic susceptibility of Fluid 1. Since the flow is irrotational ($\mathbf{v} = \nabla \phi$), integrating Equation 76 along the vertical domain yields the thermodynamic pressure p_1 . The base state pressure gradients for both fluids evaluated at the unperturbed interface are:

$$\frac{\partial p_1^0}{\partial y} = -\frac{\mu_1}{k} U - \rho_1 g \quad (77)$$

$$\frac{\partial p_2^0}{\partial y} = -\frac{\mu_2}{k} U - \rho_2 g \quad (78)$$

Following the expansion of the dynamic boundary condition at the perturbed interface $y = \lambda$, the pressure jump requires balancing the capillary forces ($p_2 - p_1 = -\sigma \alpha^2 \lambda$). Subtracting the base state components, the linearized

dynamic condition becomes:

$$p'_2 - p'_1 + \lambda \left(\frac{\partial p_2^0}{\partial y} - \frac{\partial p_1^0}{\partial y} \right) = -\sigma \alpha^2 \lambda \quad (79)$$

Substituting the base gradients (Eq. 77) and the pressure perturbations ($p'_2 = -\frac{\mu_2}{k} \phi'_2$ and $p'_1 = -\frac{\mu_1}{k} \phi'_1 + \mu_0 \chi_1 (\mathbf{H}_{01} \cdot \mathbf{H}'_1)$), Equation 79 is expanded as:

$$\frac{\mu_1}{k} \phi'_1 - \frac{\mu_2}{k} \phi'_2 - \mu_0 \chi_1 (\mathbf{H}_{01} \cdot \mathbf{H}'_1) + \lambda \left[\frac{U}{k} (\mu_1 - \mu_2) - g(\rho_2 - \rho_1) \right] = -\sigma \alpha^2 \lambda \quad (80)$$

5.5.2 Magnetic Potential Continuity and Dispersion Relation

The magnetic perturbation term $\mathbf{H}_{01} \cdot \mathbf{H}'_1$ evaluated at $y = 0$ is analytically determined by ensuring the continuity of the magnetic scalar potential and the normal magnetic flux across the deformed interface. Adapting the spatial wavenumber to α , the scalar product evaluates strictly to:

$$\mathbf{H}_{01} \cdot \mathbf{H}'_1 \Big|_{y=0} = \lambda \frac{\alpha}{\mu_{m1} + \mu_{m2}} \left[(H_{0n1}^2 - H_{0t}^2) (\mu_{m1} - \mu_{m2}) + i\mathcal{I} \right] \quad (81)$$

where $\mathcal{I}$ represents the imaginary phase shift component induced by the tangential field.

Substituting the potential functions evaluated at the boundary ($\phi'_1 = \frac{s}{\alpha} \lambda$ and $\phi'_2 = -\frac{s}{\alpha} \lambda$) and the magnetic product (Eq. 81) into the dynamic condition (Eq. 80), the interface amplitude λ factors out. Isolating the real part of the complex growth rate ($\zeta = \text{Re}(s)$) structurally proves that the principle of exchange of stabilities holds for the real components:

$$\zeta = \frac{\alpha}{\frac{\mu_1}{k} + \frac{\mu_2}{k}} \left[\frac{U}{k} (\mu_2 - \mu_1) + g(\rho_2 - \rho_1) - \sigma \alpha^2 + \mu_0 \chi_1 \alpha \frac{\mu_{m1} - \mu_{m2}}{\mu_{m1} + \mu_{m2}} (H_{0n1}^2 - H_{0t}^2) \right] \quad (82)$$

5.5.3 Generalized Nondimensionalization

To formulate the unified nondimensional parameter space, the magnetic contrast parameter is defined as $\Lambda_m = \frac{\mu_{m1} - \mu_0}{\mu_{m1} + \mu_0}$ (recalling $\mu_{m2} = \mu_0$). A new dimensionless parameter is introduced to quantify the ratio between magnetic energy and surface tension:

- $Ca_m = \frac{\mu_0 \chi_1 H_0^2 b}{\sigma}$ (Magnetic Capillar Number)

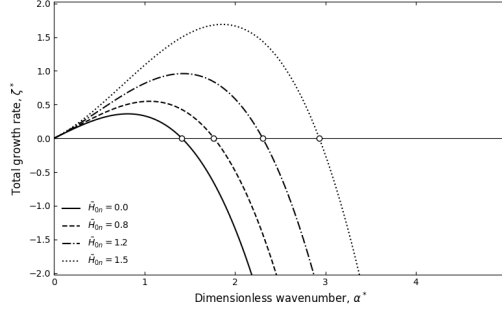


Figure 5: Enter Caption

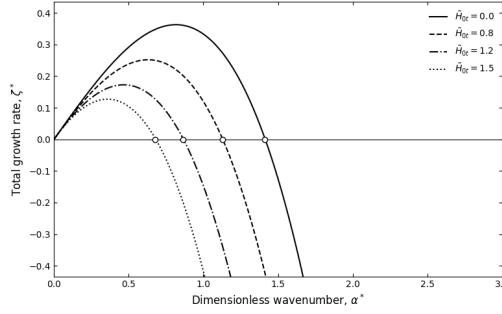


Figure 6: Enter Caption

Applying the characteristic scales established previously ($\zeta^* = \zeta b/U$, $\alpha^* = \alpha b$, $\tilde{H} = H/H_0$), and algebraically arranging the terms using the Capillary (Ca), gravitational Bond (Bo), and Darcy (Da) numbers, the exact dimensionless dispersion relation yields:

$$\zeta^*(\alpha^*) = \alpha^* \frac{M-1}{M+1} + \frac{Da}{Ca(1+M)} \alpha^* \left[Bo - \alpha^{*2} + Ca_m \Lambda_m \alpha^* (\tilde{H}_{0n}^2 - \tilde{H}_{0t}^2) \right] \quad (83)$$

This consolidated formulation explicitly maps the stabilizing/destabilizing nature of each driving force. The term α^{*2} within the brackets confirms the third-order restoring effect of capillarity ($-\alpha^{*3}$ globally), while the magnetic term introduces a second-order dependence (α^{*2} globally) that acts proportionally to the squared components of the normal and tangential fields.

Figures 5 and 6 compare the effects of applying a uniform magnetic field either normal ($\tilde{H}_{0n}$) or tangential ($\tilde{H}_{0t}$) to the fluid interface. The addition of the magnetic field introduces a quadratic term ($\propto \alpha^{*2}$) to the dispersion

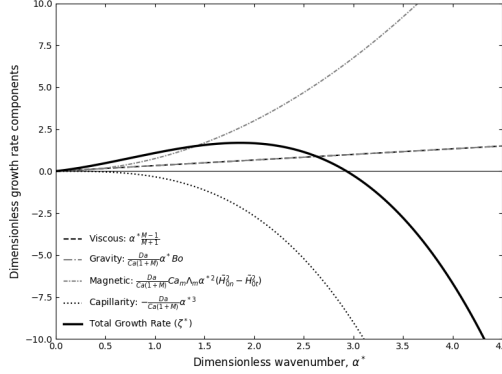


Figure 7: Enter Caption

relation, which alters the stability of the system depending entirely on the field's orientation.

When the magnetic field is normal to the interface (Figure 5), it acts as a strong destabilizing force. As $\tilde{H}_{0n}$ increases, the maximum growth rate is rapidly amplified, and the cut-off wavenumber shifts to the right. This indicates that the normal magnetic field overcomes the capillary forces, causing shorter wavelengths to become unstable and accelerating the overall deformation of the interface.

Conversely, when the magnetic field is tangential to the interface (Figure 6), it acts as a stabilizing mechanism. Increasing $\tilde{H}_{0t}$ decreases the growth rate, flattening the instability curve, and shifts the cut-off wavenumber to the left. In this configuration, the magnetic force works in the same direction as surface tension to suppress interfacial perturbations. This reduces the range of unstable waves and effectively delays the onset of the instability.

Figure 7 breaks down the individual physical forces that make up the total dispersion relation under a normal magnetic field ($\tilde{H}_{0n} > \tilde{H}_{0t}$). The total growth rate (ζ^*) is the direct sum of four components: viscosity, gravity, the magnetic force, and capillarity. As the graph shows, the viscous and gravitational forces grow linearly with the dimensionless wavenumber (α^*). However, the newly added magnetic force grows quadratically ($\propto \alpha^{*2}$). Because of this steeper curve, the magnetic term quickly outpaces the linear forces, acting as a powerful amplifier that drives the interface into a highly unstable state at intermediate wavelengths. To counteract this rapid, accelerated growth, a stronger restoring force is required. The capillarity provides this balance be-

cause it scales with the negative cube of the wavenumber ($-\alpha^*{}^3$). Although it starts small, this cubic term eventually overpowers all three destabilizing forces at higher wavenumbers. This forces the total growth rate to drop sharply below zero, ensuring the interface remains stable against microscopic perturbations.

6 Computational Methodology

6.1 Lattice Boltzmann Method and Solver Formulation

Unlike the Finite Volume Method (FVM), which discretizes the macroscopic Navier-Stokes equations and requires computationally expensive procedures to resolve pressure-velocity coupling, the Lattice Boltzmann Method (LBM) operates at the mesoscopic scale. It solves the discrete Boltzmann transport equation, recovering macroscopic hydrodynamics via the Chapman-Enskog expansion [11]. This kinetic approach offers explicit time integration, exact linear advection, and local collision rules, making it highly efficient for handling the complex boundaries inherent to porous media. [FALAR DE PORQUE É BOM PARA ESSE CASO EM ESPECIFICO]

The spatial domain is discretized into a uniform global Cartesian mesh. The velocity space at each node is quantized using the D2Q9 lattice model, which defines nine discrete velocity vectors $\mathbf{c}_i$: one rest particle, four cardinal, and four diagonal directions [12]. This specific lattice provides the necessary symmetry and isotropy to correctly recover the full Navier-Stokes stress tensor in two dimensions.

The evolution of the system is governed by the discrete Lattice Boltzmann equation employing the single-relaxation-time Bhatnagar-Gross-Krook (BGK) collision operator [13]. To accurately couple the phase-field model and introduce macroscopic forces into the mesoscopic distributions, an exact forcing term S_i [14] is utilized:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau(\phi)} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + S_i \Delta t \quad (84)$$

where f_i are the density distribution functions, $\mathbf{c}_i$ are the discrete lattice velocities, and τ is the relaxation time. Computationally, τ is linearly interpolated across the fluid interface based on the local phase fraction ϕ . This interpolation ensures a smooth transition of kinematic viscosity between the distinct fluid phases, avoiding numerical instabilities associated with sharp gradients. [FALAR SE DA PARA RECUPERAR A EQUAÇÃO DE ESCOAMENTO EM MEIO POROSO COM UM FLUIDO MULTIFASICO]

6.2 Cahn-Hilliard Solver: Spatial Discretization and Temporal Substepping

The order parameter ϕ whose theoretical formulation was established in the Phase-Field Modeling section is evolved by a dedicated finite-difference module that runs alongside the LBM hydrodynamic kernel. Because the advective Cahn-Hilliard equation

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = M \nabla^2 \mu_c, \quad \mu_c = 4\beta \phi(\phi^2 - 1) - \kappa \nabla^2 \phi, \quad (85)$$

is fourth-order in ϕ , a one-shot discretization would require a wide biharmonic stencil. Instead, the implementation splits the update into two coupled second-order passes — first computing the chemical potential field μ_c over the entire domain, then advancing ϕ using $\nabla^2 \mu_c$ — which reduces every stencil to five points and makes the algorithm naturally local.

6.2.1 Five-Point Isotropic Stencil

All spatial derivatives are evaluated on the same uniform Cartesian grid used by the LBM kernel, with lattice spacing $\Delta x = 1$ in lattice units. The Laplacian of any scalar field ψ is approximated by the standard five-point stencil:

$$\nabla^2 \psi(y, x) \approx \psi(y, x+1) + \psi(y, x-1) + \psi(y+1, x) + \psi(y-1, x) - 4\psi(y, x). \quad (86)$$

The first spatial derivatives entering the advective term $\mathbf{u} \cdot \nabla \phi$ are computed using second-order centred differences, e.g. $\partial_x \phi \approx [\phi(y, x+1) - \phi(y, x-1)]/2$. The five-point Laplacian carries a small leading-order anisotropy, but this is acceptable here because the Cahn-Hilliard relaxation acts on a thermodynamic timescale much shorter than the hydrodynamic timescale, smoothing out grid-induced artefacts. By contrast, the macroscopic force computation in the LBM kernel uses the more demanding 9-point isotropic stencil, since spurious anisotropic currents at the interface would directly contaminate the momentum balance.

6.2.2 Boundary Conditions

The boundary treatment of the order parameter is fully consistent with the boundary conditions imposed by the LBM kernel. The transverse boundaries

($y = 0$ and $y = N_y - 1$) are always treated as periodic, mirroring the spanwise periodicity of the LBM mesh. The longitudinal boundaries depend on the simulation mode:

- **Inlet ($x = 0$, injection mode):** a Dirichlet condition $\phi = +1$ is enforced, representing the continuous injection of the pure invading phase. The condition is reapplied both before computing μ_c and after the temporal update, to prevent the explicit Euler step from disturbing the boundary value through the centred-difference advective stencil.
- **Outlet ($x = N_x - 1$):** a homogeneous Neumann condition $\partial_x \phi = 0$ is enforced by copying the value from the adjacent internal node, $\phi_{N_x-1} = \phi_{N_x-2}$. This allows the interface to be advected smoothly out of the computational domain without spurious reflection.
- **Chemical potential μ_c at both x -boundaries:** homogeneous Neumann ($\partial_x \mu_c = 0$). Physically, this nullifies the diffusive flux $M \nabla \mu_c$ across the longitudinal boundaries — at the inlet, the phase composition is dictated by the injection Dirichlet condition, not by diffusion; at the outlet, the only allowed mass flux is the advective sweep by the velocity field.

For fully periodic configurations (e.g. spatially periodic linear-stability test cases), both x -boundaries are treated periodically through modular indexing, identical to the y treatment.

6.2.3 Explicit Euler Update and Numerical Safeguards

After the chemical potential field has been finalised, the order parameter is advanced through an explicit first-order Euler scheme:

$$\phi^{n+1}(y, x) = \phi^n(y, x) + \Delta t_{CH} [M \nabla^2 \mu_c - (u_x \partial_x \phi + u_y \partial_y \phi)]. \quad (87)$$

The velocity field (u_x, u_y) here is the physical velocity recovered from the Guo forcing scheme described below. To prevent unphysical overshoots of the order parameter beyond the bulk equilibrium values $\phi = \pm 1$ — overshoots that the quartic bulk term of $\mathcal{F}$ can amplify into catastrophic numerical blow-up — the updated values are clamped to the physical interval $\phi^{n+1} \in [-1, +1]$ before being stored. In well-resolved simulations (sufficient interface thickness and properly calibrated mobility), the clamp is essentially never

active and the associated mass error is negligible; the clamp is retained as a numerical safety net for transient instabilities, e.g. during the start-up phase or near very high mobility-ratio interfaces.

The update is implemented through a double-buffer pattern (`phi_in`, `phi_out`) so that the right-hand side at each grid point is evaluated using the field ϕ^n everywhere, never with partially updated values from earlier in the sweep. The buffer pointers are swapped externally between substeps, avoiding any in-place memory hazard.

6.2.4 Temporal Substepping and CFL Constraint

The Cahn-Hilliard equation is fourth-order in space, so the explicit-Euler stability bound scales as $(\Delta x)^4$:

$$\Delta t_{CH} \lesssim \frac{(\Delta x)^4}{4 M \kappa}, \quad (88)$$

which is considerably more restrictive than the convective CFL of the LBM kernel ($\Delta t_{LBM} \lesssim \Delta x/c_s$). To preserve the natural unit time step of the LBM while honouring the Cahn-Hilliard stability bound, the solver is invoked N_{sub} times per LBM step, with reduced substep $\Delta t_{CH} = \Delta t_{LBM}/N_{sub}$. The number of substeps is chosen at runtime such that the diffusive CFL is satisfied with a safety margin, conventionally targeting $M \kappa \Delta t_{CH}/(\Delta x)^4 \leq 0.1$. For the simulations presented in this work, typical values fall in the range $N_{sub} \in [5, 20]$, with larger values required when σ (and therefore the product $\kappa\beta$) is increased to suppress numerical mixing at high mobility ratio.

6.2.5 Parallelization and Coupling

The substep routine is compiled by Numba with `@jit(parallel=True, cache=True)`, and the outer loop over transverse rows (y) is parallelised using `prange`. Because the five-point stencil is local and the temporal update writes only to `phi_out`, there are no inter-row data dependencies and the parallel reduction is embarrassingly parallel. The chemical-potential computation and the temporal update are kept in separate parallel passes because the second pass reads μ_c at the four nearest neighbours and therefore requires the μ_c field to be fully populated across all rows before any value of ϕ^{n+1} is computed. This two-phase structure is the algorithmic image of the second-order operator splitting of the original fourth-order equation.

At the end of each LBM step, the updated phase field ϕ is consumed by three physically distinct couplings in the next iteration: the heterogeneous magnetic susceptibility $\chi(\phi)$ used by the Poisson solver, the local permeability $K(\phi)$ used by the Guo forcing scheme, and the interpolated kinematic viscosity $\nu(\phi)$ entering the BGK relaxation time. The capillary body force $\mathbf{F}_s = \mu_c \nabla \phi$ is reassembled inside the LBM kernel from the latest ϕ field rather than being passed in from the Cahn-Hilliard module, so that all three macroscopic forces ($\mathbf{F}_s$, $\mathbf{F}_m$, and the Darcy drag) act on the same instantaneous phase configuration.

6.3 External Forcing: Interfacial and Magnetic Couplings

The total macroscopic body force $\mathbf{F} = \mathbf{F}_s + \mathbf{F}_m$ acts as a source term for the momentum update at each discrete time step.

The isotropic capillary force (Korteweg stress) is recovered directly from the chemical potential gradient of the Cahn-Hilliard phase-field model [15]:

$$\mathbf{F}_s = \mu \nabla \phi = [4\beta\phi(\phi^2 - 1) - \kappa \nabla^2 \phi] \nabla \phi \quad (89)$$

The magnetic force (Kelvin force) arises from heterogeneities in the magnetic susceptibility $\chi(\phi)$ subjected to an external scalar potential field ψ . To suppress grid-dependent truncation errors in the numerical implementation, this force is computed via tensorial decomposition:

$$\mathbf{F}_m = \chi \left(\frac{\partial \psi}{\partial x} \nabla \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y} \nabla \frac{\partial \psi}{\partial y} \right) \quad (90)$$

All spatial derivatives and Laplacians required to compute these interfacial and magnetic forces are evaluated using 9-point isotropic finite difference schemes. This approach mitigates the intrinsic geometric anisotropy of the Cartesian grid and improves numerical stability near moving interfaces. [CITAR E EXPLICAR A FORÇA DE KELVIN DE ACORDO COM ARTIGO DE FRANCISCO RICARDO DA CUNHA E RAFAEL GABLER CONTIJO. O CASO DESSES ARTIGOS DELE USA O POTENCIAL MAGNETICO TBM?. EXPLIQUE MELHOR PORQUE ESSE METODO REDUZ OS ERROS DE TRUNCAMENTO E POR QUE MITIGA A ANISOTROPIA DA GRID]

6.4 Darcy Resistance and Kinematic Recovery (Guo Scheme)

To incorporate the Darcy-Brinkman resistance tensor representing a porous medium with permeability K , the physical fluid velocity $\mathbf{u}_{phys}$ must be decoupled from the raw first-order statistical moment. The implementation adopts the Guo forcing scheme [14], which implicitly recovers the true physical velocity by analytically solving for the drag parameter $\sigma = \nu/K$:

$$\mathbf{u}_{phys} = \frac{\sum f_i \mathbf{c}_i + \frac{\Delta t}{2}(\mathbf{F}_s + \mathbf{F}_m)}{1 + \frac{\Delta t}{2}\sigma} \quad (91)$$

Consequently, the effective total force used to compute the discrete source term S_i in the evolution equation is dynamically adjusted by subtracting the viscous pressure drop induced by the solid matrix: $\mathbf{F}_{total} = (\mathbf{F}_s + \mathbf{F}_m) - \sigma \rho \mathbf{u}_{phys}$.

6.5 Boundary Conditions

The streaming step advects post-collision populations to adjacent nodes. At the longitudinal solid boundaries ($y = 0$ and $y = N_y - 1$), the no-slip condition is enforced via the half-way bounce-back scheme [16]. Algorithmically, this reflects the incoming distribution functions strictly back to their origin node ($\mathbf{c}_{\bar{i}} = -\mathbf{c}_i$), ensuring exact mass conservation at the walls.

At the transverse open boundaries, asymmetric conditions are implemented to simulate a semi-infinite flow-driven channel:

- **Inlet** ($x = 0$): A constant velocity Dirichlet condition is applied. Missing incoming populations are reconstructed using the equilibrium distribution f_i^{eq} defined by the prescribed velocity $\mathbf{u} = (U_{in}, 0)$. To prevent unphysical pressure build-up and maintain consistency with the internal Darcy resistance field, the local macroscopic density is not hard-coded; instead, it is backward-extrapolated from the first internal fluid node ($\rho = \rho_{x=1}$).
- **Outlet** ($x = N_x - 1$): A Neumann-type convective boundary condition is applied to allow the smooth exit of interfacial perturbations. The code implements a second-order spatial extrapolation for the populations ($f_i^{N_x-1} = 2f_i^{N_x-2} - f_i^{N_x-3}$). This numerical approach effectively

nullifies the second derivatives at the boundary, mitigating spurious mass and acoustic reflections back into the computational domain.

6.6 Algorithm Structure and Main Loop

The computational implementation strictly follows the standard LBM algorithm structure. After initializing the macroscopic fields $(\rho, \mathbf{u}, \phi)$ and establishing the initial equilibrium distributions, the main temporal loop executes the following sequential steps at each iteration:

1. **Macroscopic and Forcing Calculation:** The interfacial variables and external forces $(\mathbf{F}_s, \mathbf{F}_m)$ are evaluated using spatial finite differences. The physical velocity $\mathbf{u}_{phys}$ is subsequently recovered.
2. **Collision Step:** The local equilibrium distribution f_i^{eq} and source term S_i are computed. The distributions f_i are then relaxed locally according to the BGK operator.
3. **Streaming Step:** The post-collision populations are advected linearly to their respective neighboring nodes along the $\mathbf{c}_i$ directions.
4. **Boundary Conditions:** The bounce-back and open boundary rules are applied to reconstruct unknown incoming populations at the domain edges.
5. **Macroscopic Update:** The fluid density ρ and phase fraction ϕ are updated via the zeroth-order moments of their respective distributions, advancing the system to $t + \Delta t$.

6.7 Code Validation: 2D Poiseuille Flow

To verify the accuracy of the implemented LBM collision operators and boundary conditions, a fundamental Poiseuille flow in a two-dimensional channel was simulated. The computational domain was discretized into $N_x \times N_y = 200 \times 50$ lattice nodes. The flow is kinematically driven by a constant inlet velocity condition of $U_{in} = 0.01$ lattice units (lu) enforced at $x = 0$, while the zero-gradient convective boundary condition is applied at the outlet ($x = 199$). No-slip conditions at the upper and lower rigid walls are recovered via the half-way bounce-back rule [EXPLICAR MELHOR O

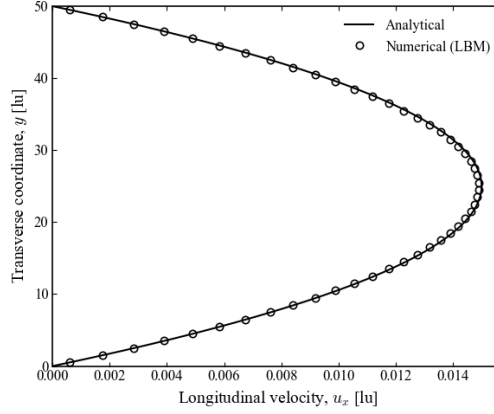


Figura 8: Enter Caption

QUE SÃO O HALF-WAY E O BOUNCE-BACK E PORQUE SE PODE USAR ISSO, ALEM DE CITAR ARITGO QUE DESENVOLVEU ISSO]. [FALAR QUE O CASO SIMULADO ASSIM ESTA EM UMA BRANCH DO GITHUB + PASSAR LINK]

The theoretical velocity profile for a fully developed, steady, incompressible channel flow is given by the parabolic equation [CITAR ARTIGO]:

$$u_x(y) = \frac{4U_{max}}{H^2}y(H - y) \quad (92)$$

where H is the physical channel width and $U_{max} = 1.5U_{in}$ is the theoretical centerline velocity.

A critical aspect of comparing the numerical LBM results with the analytical continuous formulation is the spatial coordinate correction. Because the half-way bounce-back scheme algorithmically locates the physical solid wall exactly halfway between the boundary fluid node and the external ghost node, the numerical transverse coordinates must be shifted by $\Delta y = 0.5$ lu, such that $y_{phys} = y_{lattice} + 0.5$. Consequently, the effective physical channel width is exactly $H = N_y$.

The numerical longitudinal velocity profile (u_x) was extracted at the transversal section $x = 190$ to guarantee that the hydrodynamic analysis occurs within the fully developed flow region. The numerical solution exhibited excellent agreement with the analytical prediction, as shown in Figure ??.

The quantitative convergence analysis validates the incompressibility assumption and the robustness of the boundary schemes. For the prescribed inlet velocity, the measured mean velocity in the evaluation section was 0.009942 lu. The local divergence between the simulated and exact fields resulted in a relative L_2 norm error of 6.0272×10^{-3} . Furthermore, the absolute deviations remained strictly bounded, yielding a Mean Absolute Error (MAE) of 5.9861×10^{-5} lu and a Root Mean Square Error (RMSE) of 6.6025×10^{-5} lu, confirming the second-order spatial accuracy characteristic of the standard D2Q9 LBM solver.

7 Darcy-Brinkman Regime and Permeability Validation

7.1 Guo Forcing Scheme and Kinematic Synchronization

In porous media simulations, the solid matrix induces a macroscopic drag force $\mathbf{F}_d = -\frac{\nu}{K}\rho\mathbf{u}$. Incorporating this velocity-dependent resistance directly into the standard Lattice Boltzmann equation introduces numerical instability and breaks the second-order spatial accuracy of the collision operator.

To resolve this, the implementation utilizes the exact forcing scheme proposed by Guo et al. [14]. This method analytically decouples the true physical velocity $\mathbf{u}$ from the macroscopic force computation by solving the implicit momentum balance:

$$\mathbf{u} = \frac{\sum f_i \mathbf{c}_i + \frac{\Delta t}{2} \mathbf{F}_{ext}}{\rho \left(1 + \frac{\Delta t}{2} \frac{\nu}{K}\right)} \quad (93)$$

where $\mathbf{F}_{ext}$ encapsulates all explicit external body forces (e.g., capillary or magnetic forces) and K is the intrinsic permeability.

This formulation guarantees algorithmic synchronization with normal free-flow regions. In zones without porous structures ($K \rightarrow \infty$), the structural resistance denominator converges to unity ($\sigma = \nu/K \rightarrow 0$). Consequently, the system recovers standard Navier-Stokes hydrodynamics smoothly and continuously, eliminating the need for conditional logic (branching) at fluid-porous interfaces. [ESSA PARTE ESTÁ REDUNDANTE EM PARTES COM O

7.2 Data Extraction and K Recovery Methodology

The numerical validation of the porous resistance was conducted in a 200×50 lattice domain. The flow was driven by a constant kinematic inlet ($U_{in} = 0.01$ lu) and constrained by a strict thermodynamic Dirichlet boundary at the outlet ($\rho_{out} = 1.0$). [FALAR QUE O CASO SIMULADO ASSIM ESTA EM UMA BRANCH DO GITHUB + PASSAR LINK]

To rigorously evaluate the permeability recovery, it is mandatory to eliminate spurious kinetic artifacts generated by non-equilibrium population reconstructions at the open boundaries. Therefore, the hydrodynamic variables were extracted exclusively from the channel centerline ($y = N_y/2$) within a spatially truncated, fully developed “bulk region” defined between $x = 0.25N_x$ and $x = 0.75N_x$.

The thermodynamic pressure P was computed directly from the local density via the LBM isothermal equation of state ($P = c_s^2 \rho$). For a homogeneous porous medium, Darcy’s Law dictates that the longitudinal pressure profile must exhibit a strictly linear drop. The macroscopic pressure gradient ∇P was extracted numerically via a linear least-squares regression applied to the centerline pressure nodes within the bulk region.

Subsequently, the algorithm’s recovered numerical permeability (K_{num}) was calculated by inverting Darcy’s equation:

$$K_{num} = \frac{\nu \bar{\rho} \bar{u}_x}{-\nabla P} \quad (94)$$

where $\bar{\rho}$ and $\bar{u}_x$ are the spatially averaged density and longitudinal velocity across the bulk region.

7.3 Validation Results

The numerical framework was evaluated against analytical solutions for Brinkman flow in a channel bounded by parallel no-slip walls [17]:

$$u_x(y) = U_{darcy} \left[1 - \frac{\cosh\left(\frac{y-H/2}{\sqrt{K_0}}\right)}{\cosh\left(\frac{H}{2\sqrt{K_0}}\right)} \right] \quad \text{where} \quad U_{darcy} = -\frac{\nabla P \cdot K_0}{\nu \bar{\rho}} \quad (95)$$

As depicted in Figure 9, for a baseline test with imposed permeability $K_0 = 10.0$ lu, the LBM solver accurately captured the linear thermodynamic pressure decay (yielding a recovered $K_{num} \approx 9.99$ lu). The extracted

transverse velocity profile perfectly reproduced the analytical plug-flow characteristic induced by the Brinkman viscous resistance.

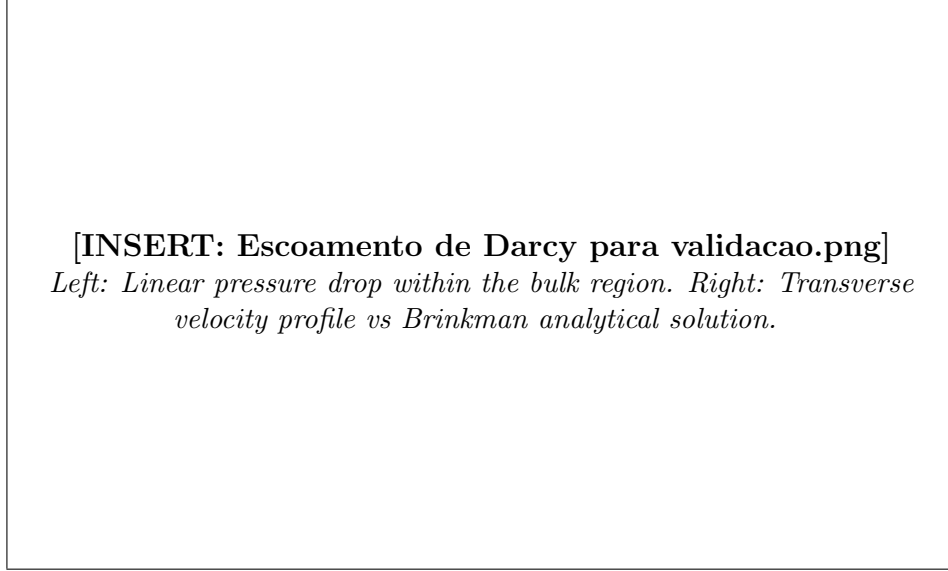


Figura 9: Hydrodynamic validation of the internal resistance field for $K_0 = 10.0$ lu.

To map the stability limits of the forcing scheme, an automated parametric sweep was conducted for $K_0 \in [0.5, 20.0]$ lu. Figure 10 confirms an ideal parity between the imposed K_0 and the numerical K_{num} .

The relative computational error remained strictly bounded below 0.8% for all scenarios. Accuracy maximizes (error $\approx 0.1\%$) at lower permeabilities, where the solid matrix drag dominates viscous diffusion. This fully validates the thermodynamic consistency of the LBM implementation and the spatial accuracy of the Guo forcing formulation.

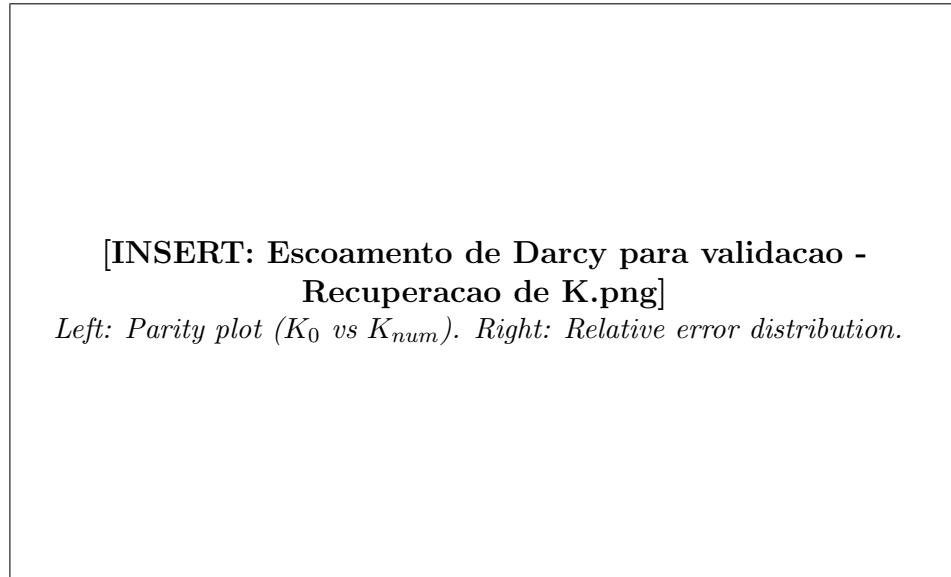


Figura 10: Permeability recovery performance and relative computational error across different porous regimes.

8 Results

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